

Calculus I Lesson 11

$$1) f(x) = \frac{1}{x-2}$$

$$a) \lim_{x \rightarrow 2^-} f(x) = \underline{\quad ? \quad}$$

$$b) \lim_{x \rightarrow 2^+} f(x) = \underline{\quad ? \quad}$$

$$c) \lim_{x \rightarrow 2} f(x) = \underline{\quad ? \quad}$$

$$2) f(x) = \frac{3}{x^2 - 4}$$

$$\text{a) } \lim_{x \rightarrow 2^-} f(x) = \underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$$

$$\text{b) } \lim_{x \rightarrow 2^+} f(x) = \underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$$

$$\text{c) } \lim_{x \rightarrow 2} f(x) = \underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$$

$$\text{d) } \lim_{x \rightarrow -2^-} f(x) = \underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$$

$$\text{e) } \lim_{x \rightarrow -2^+} f(x) = \underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$$

$$\text{f) } \lim_{x \rightarrow -2} f(x) = \underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$$

$$3) f(x) = \frac{4}{(x-1)^2}$$

$$\text{a) } \lim_{x \rightarrow 1^-} f(x) = \underline{\hspace{2cm} ? \hspace{2cm}}$$

$$\text{b) } \lim_{x \rightarrow 1^+} f(x) = \underline{\hspace{2cm} ? \hspace{2cm}}$$

$$\text{c) } \lim_{x \rightarrow 1} f(x) = \underline{\hspace{2cm} ? \hspace{2cm}}$$

$$4) \ f(x) = \left(2 + \frac{3}{x-2} \right)$$

$$\text{a) } \lim_{x \rightarrow 2^-} f(x) = \underline{\hspace{2cm}}?$$

$$\text{b) } \lim_{x \rightarrow 2^+} f(x) = \underline{\hspace{2cm}}?$$

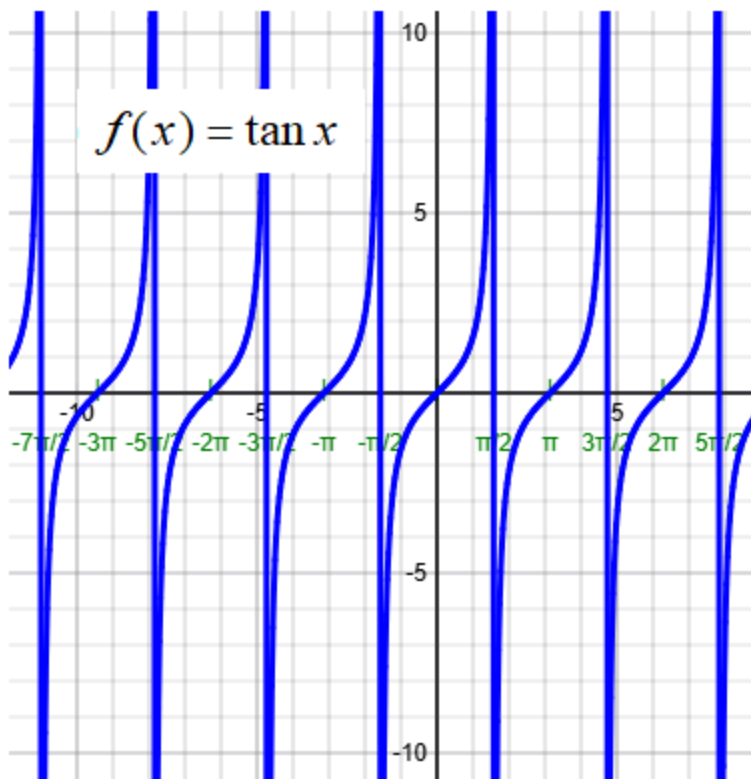
$$\text{c) } \lim_{x \rightarrow 2} f(x) = \underline{\hspace{2cm}}?$$

5) $f(x) = \tan x$

a) $\lim_{x \rightarrow \pi/2^-} f(x) = \underline{\hspace{2cm}} ?$

b) $\lim_{x \rightarrow \pi/2^+} f(x) = \underline{\hspace{2cm}} ?$

c) $\lim_{x \rightarrow \pi/2} f(x) = \underline{\hspace{2cm}} ?$



6) $f(x) = \cot x$

a) $\lim_{x \rightarrow \pi^-} f(x) = \underline{\hspace{2cm}} ?$

b) $\lim_{x \rightarrow \pi^+} f(x) = \underline{\hspace{2cm}} ?$

c) $\lim_{x \rightarrow \pi/2} f(x) = \underline{\hspace{2cm}} ?$

